

Effectiveness of alcohol-based hand rubs for removal of *Clostridium difficile* spores from hands.

BACKGROUND: Alcohol-based hand rubs (ABHRs) are an effective means of decreasing the transmission of bacterial pathogens. Alcohol is not effective against *Clostridium difficile* spores. We examined the retention of *C. difficile* spores on the hands of volunteers after ABHR use and the subsequent transfer of these spores through physical contact. **METHODS:** Nontoxigenic *C. difficile* spores were spread on the bare palms of 10 volunteers. Use of 3 ABHRs and chlorhexidine soap-and-water washing were compared with plain water rubbing alone for removal of *C. difficile* spores. Palmar cultures were performed before and after hand decontamination by means of a plate stamping method. Transferability of *C. difficile* after application of ABHR was tested by having each volunteer shake hands with an uninoculated volunteer. **RESULTS:** Plain water rubbing reduced palmar culture counts by a mean ($\pm$ standard deviation [SD]) of 1.57 ± 0.11 log₁₀ colony-forming units (CFU) per cm², and this value was set as the zero point for the other products. Compared with water washing, chlorhexidine soap washing reduced spore counts by a mean ($\pm$ SD) of 0.89 ± 0.34 log₁₀ CFU per cm²; among the ABHRs, Isagel accounted for a reduction of 0.11 ± 0.20 log₁₀ CFU per cm² ($P = .005$), Endure for a reduction of 0.37 ± 0.42 log₁₀ CFU per cm² ($P = .010$), and Purell for a reduction of 0.14 ± 0.33 log₁₀ CFU per cm² ($P = .005$). There were no statistically significant differences between the reductions achieved by the ABHRs; only Endure had a reduction statistically different from that for water control rubbing ($P = .040$). After ABHR use, handshaking transferred a mean of 30% of the residual *C. difficile* spores to the hands of recipients. **CONCLUSIONS:** Hand washing with soap and water is significantly more effective at removing *C. difficile* spores from the hands of volunteers than are ABHRs. Residual spores are readily transferred by a handshake after use of ABHR.